STOCHASTIC METHODS IN FINANCE
STAT0013
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Suggested Solutions to Exercises 8

1. Recall that $X_t = at + \sigma B_t$. Notice that $(dX_t)^2 = \sigma^2 dt$.

   (a) Let $F_t = e^{-X_t}$. From Itô's formula, we have

   $$\begin{aligned} dF_t &= -e^{-X_t} dX_t + \tfrac{1}{2} e^{-X_t} (dX_t)^2 \\ &= -F_t(adt + \sigma dB_t) + \tfrac{\sigma^2}{2} F_t dt. \end{aligned}$$

   Thus,

   $$dF_t = F_t(\tfrac{\sigma^2}{2} - a)dt - \sigma F_t dB_t.$$

   So, F_t is a Geometric Brownian motion.

   (b) Let $G_t = \cos(e^{X_t})$. From Itô's formula, we have

   $$dG_t = -\sin(e^{X_t})e^{X_t} dX_t + \tfrac{1}{2}\big[-\cos(e^{X_t})e^{2X_t} - \sin(e^{X_t})e^{X_t} \big] (dX_t)^2.$$

   Thus,

   $$\begin{aligned} dG_t = \big[-a\sin(e^{X_t})e^{X_t} - \tfrac{\sigma^2}{2}\cos(e^{X_t})e^{2X_t} - \tfrac{\sigma^2}{2}\sin(e^{X_t})e^{X_t} \big] dt \\ - \big[\sigma \sin(e^{X_t})e^{X_t} \big] dB_t. \end{aligned}$$

   It is fine to stop here, but if one wants to write the SDE more in terms of G_t, notice that $\sin(e^x) = \sqrt{1-G_t^2}$. So, one can re-write the SDE as

   $$\begin{aligned} dG_t = \big[-a\sqrt{1-G_t^2}\, e^{X_t} - \tfrac{\sigma^2}{2} G_t\, e^{2X_t} - \tfrac{\sigma^2}{2} \sqrt{1-G_t^2}\, e^{X_t} \big] dt \\ - \big[\sigma \sqrt{1-G_t^2}\, e^{X_t} \big] dB_t. \end{aligned}$$

2. Note that $(dS_t)^2 = \sigma^2 S_t^2\, dt$.

(a) $X_t = \log(S_t)$.
From Itô's formula, we have

$$dX_t = \frac{1}{S_t} dS_t - \frac{1}{2}\frac{1}{S_t^2}(dS_t)^2 \quad \Longrightarrow$$

$$dX_t = \big(\alpha - \delta - \tfrac{\sigma^2}{2}\big)dt + \sigma\, dB_t$$

(b) $X_t = \sqrt{S_t}$.
From Itô's formula, we have

$$\begin{aligned} dX_t &= \tfrac{1}{2}S_t^{-1/2}dS_t + \tfrac{1}{2}(-\tfrac{1}{4})S_t^{-3/2}(dS_t)^2 \quad \Longrightarrow \\ dX_t &= \big(\tfrac{\alpha-\delta}{2}S_t^{1/2} - \tfrac{\sigma^2}{8}S_t^{1/2}\big)dt + \tfrac{1}{2}\sigma\, S_t^{1/2}\, dB_t \Longrightarrow \\ dX_t &= \big(\tfrac{\alpha-\delta}{2} - \tfrac{\sigma^2}{8}\big)X_t dt + \tfrac{1}{2}\sigma\, X_t\, dB_t \end{aligned}$$

(c) $X_t = e^{t/2}\sin(S_t)$.
Let $x = e^{t/2}\sin(s)$. We find the partial derivatives

$$\frac{\partial x}{\partial s} = e^{t/2}\cos(s); \qquad \frac{\partial^2 x}{\partial s^2} = -e^{t/2}\sin(s) = -x,$$
$$\frac{\partial x}{\partial t} = \tfrac{1}{2}e^{t/2}\sin(s) = \tfrac{x}{2}.$$

From Itô's formula, we have

$$\begin{aligned} dX_t = \big[\, & e^{t/2}(\alpha-\delta)\cos(S_t)S_t + \tfrac{1}{2}X_t - \tfrac{\sigma^2}{2}X_t S_t^2 \,\big]dt \\ & + \sigma\, e^{t/2}\cos(S_t)S_t\, dB_t \end{aligned}$$

Notice that $(e^{t/2}\sin(s))^2 + (e^{t/2}\cos(s))^2 = e^t$. Thus,

$$\frac{\partial x}{\partial s} = e^{t/2}\cos(s) = \sqrt{e^t - x^2}.$$

So, we obtain the expression

$$\begin{aligned} dX_t = \big[\, & \sqrt{e^t - X_t^2}\,(\alpha-\delta)S_t + \tfrac{1}{2}X_t - \tfrac{\sigma^2}{2}X_t S_t^2 \,\big]dt \\ & + \sigma\,\sqrt{e^t - X_t^2}\, S_t\, dB_t \end{aligned}$$

3. (a) Recall that the forward price at time t is related to S_t by

$$F(S_t, t) = S_t e^{r(T-t)},$$

where T is the maturity time.
We calculate the partial derivatives

$$\frac{\partial F}{\partial S} = e^{r(T-t)}; \qquad \frac{\partial^2 F}{\partial S^2} = 0; \qquad \frac{\partial x}{\partial t} = -rSe^{r(T-t)} = -rF.$$

Thus, from Itô's formula, we have that

$$dF_t = e^{r(T-t)}\, dS_t - rF_t dt \qquad \Longrightarrow$$
$$dF_t = (\mu - r)F_t dt + \sigma F_t dB_t$$

(b) F_t is a Geometric BM, so we know that, for given t_0, and $t > t_0$:

$$F_t = F_{t_0}\, e^{(\mu - r - \frac{\sigma^2}{2})(t-t_0) + \sigma(B_t - B_{t_0})}.$$

where $F_{t_0} = S_{t_0} e^{r(T-t_0)}$. Thus, we have, for $t = t_1$:

$$F(S_{t_1}, t_1) = S_{t_0} e^{r(T-t_0)} e^{(\mu - r - \frac{\sigma^2}{2})(t_1 - t_0) + \sigma(B_{t_1} - B_{t_0})}.$$

We can see that $F(S_{t_1}, t_1)$ follows a log-normal distribution. In particular,

$$\log F(S_{t_1}, t_1)$$
$$\sim N\Big(\log S_{t_0} + r(T - t_0) + (\mu - r - \tfrac{\sigma^2}{2})(t_1 - t_0), \sigma^2(t_1 - t_0)\Big).$$

4. Let $X_t = \sqrt{V_t}$. Then, we are given from the statement of the problem that

$$dX_t = (2 - 4X_t)dt + X_t^2 dB_t$$

We are interested in $V_t = X_t^2$. Application of Itô's formula gives

$$dV_t = 2X_t dX_t + (dX_t)^2 \quad \Longrightarrow$$
$$dV_t = \big[\, 2X_t(2 - 4X_t) + X_t^4 \,\big]\, dt + 2X_t^3 dB_t \quad \Longrightarrow$$
$$dV_t = \big[\, 4\sqrt{V_t} - 8V_t + V_t^2 \,\big]\, dt + 2V_t^{3/2} dB_t.$$

5. The best option here is to apply some “back-engineering”. It is not difficult to understand that X_t must be of the form

$$X_t = S_t^3 + C$$

for some constant C, since use of Itô’s formula for such X_t would indeed give (recall that $(dS_t)^2 = dt$)

$$dX_t = 3S_t^2 dS_t + 3S_t dt \quad \Longrightarrow$$

$$dX_t = 3S_t dt + 3S_t^2 dB_t$$

To find C, we use the boundary condition $X_0 = 0 = C$, so $C = 0$. Thus, we have found that

$$X_t = S_t^3.$$

6. Same idea as in Question 5, only a bit more difficult. It is not difficult to see that X_t must be of the form

$$X_t = e^{S_t}\sin(S_t) + C$$

for some constant C, since use of Itô’s formula for such X_t would indeed give (recall that $(dS_t)^2 = dt$)

$$dX_t = \left[\, e^{S_t}\sin(S_t) + e^{S_t}\cos(S_t)\,\right] dS_t + \left[\, e^{S_t}\cos(S_t)\,\right] dt$$

as required. To find the constant C, we use the boundary condition $X_0 = C = 5$, so we have obtained that

$$X_t = e^{S_t}\sin(S_t) + 5.$$

7. We have that

$$S_3 - S_0 = 3 + 2B_3.$$

Also,

$$S_5 - S_3 = 10 + (B_5 - B_3).$$

Combining the above two expressions gives

$$\begin{aligned} S_5 &= S_0 + (S_3 - S_0) + (S_5 - S_3) \\ &= 14 + 2B_3 + (B_5 - B_3) \end{aligned}$$

Thus, since the increments of BM are independent, we have that

$$S_5 \sim N(14, 14).$$

8. We have that $Y_t = S_t^3$ Using Itô's formula, we obtain

$$\begin{aligned}
dY_t &= 3S_t^2 dS_t + 3S_t(dS_t)^2 \quad \Longrightarrow \\
dY_t &= 3S_t^2 dS_t + 3\sigma^2 S_t^3 d_t \quad \Longrightarrow \\
dY_t &= (3\mu S_t^3 + 3\sigma^2 S_t^3)dt + 3\sigma S_t^3 dB_t \quad \Longrightarrow \\
dY_t &= (3\mu + 3\sigma^2)Y_t dt + 3\sigma Y_t dB_t
\end{aligned}$$

Thus, Y_t is a Geometric BM, and we have

$$Y_t = Y_0 \exp\left\{(3\mu + 3\sigma^2 - \tfrac{9\sigma^2}{2})t + 3\sigma B_t\right\}$$

Since, $S_0 = k$, we get that $Y_0 = S_0^3 = k^3$. Thus, we have a log-normal distribution for Y_t, in particular

$$\log Y_t \sim N\Big(3\log k + 3(\mu - \tfrac{\sigma^2}{2})t,\, 9\sigma^2 t\Big).$$

9. Since S_t is a Geometric BM, we can obtain the solution

$$S_t = S_0 \exp\left\{(\mu - \tfrac{\sigma^2}{2})t + \sigma B_t\right\}.$$

The riskless asset will be worth $S_0 e^{rT}$ at time T. We want to find the probability

$$\begin{aligned}
\mathrm{P}\,[\, S_T > S_0 e^{rT}\,] &= \mathrm{P}\left[\, S_0 \exp\left\{(\mu - \tfrac{\sigma^2}{2})T + \sigma B_T\right\} > S_0 e^{rT}\,\right] \\
&= \mathrm{P}\,[\,(\mu - \tfrac{\sigma^2}{2})T + \sigma B_T > rT\,] \\
&= \mathrm{P}\left[\, B_T > \tfrac{rT - (\mu - \frac{\sigma^2}{2})T}{\sigma}\,\right] \\
&= \mathrm{P}\left[\, Z > \tfrac{rT - (\mu - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\,\right] \\
&= 1 - N\Big(\tfrac{rT - (\mu - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\Big)
\end{aligned}$$

where $N(\cdot)$ denotes the cdf of the standard normal distribution.

To hedge our portfolio, in the case where the stock underperforms, we choose e.g. a position which is:
Long 1 put option with strike price $S_0 e^{rT}$ & Long 1 share.
Thus, the value of our position at time T is:
i) S_T, if $S_T > S_0 e^{rT}$; ii) $S_0 e^{rT}$, if $S_T < S_0 e^{rT}$.